

DeepFrailty and machine learning modelling for predicting under-five survival in southern Africa

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Abstract

Aim: Child health outcomes are central to Sustainable Development Goals (SDGs) 3 and 10. This study aims to predict survival probabilities for children under-five years of age in southern Africa.

Methods: We use the most recent Demographic and Health Survey (DHS) data from Malawi, South Africa, Zambia, and Zimbabwe. Machine learning and deep learning techniques for survival, including random survival forest (RSF), DeepHit, and DeepFrailty, are applied to predict survival factors. Unobserved community-level heterogeneity is accounted for by including cluster identifiers in the DeepFrailty model. Model (s) performance is evaluated using concordance indices (C-index) and Integrated Brier Scores (IBS).

Results: The RSF model consistently shows the highest predictive performance (C-index: 0.8890-0.9458; IBS: 0.0062-0.0097), demonstrating strong reliability across countries. The DeepHit model, while exhibiting slightly lower performance (C-index: 0.5889-0.8943; IBS: 0.0085-0.0140), showed comparable results in Malawi and Zambia. The DeepFrailty model, though lower in predictive accuracy (C-index: 0.8385-0.9159; IBS: 0.0164-0.0212), adds interpretive value by capturing unobserved, community-level heterogeneity. The key predictors of child survival include family structure and maternal care, with

“births in the past year,” “total number of children born,” “currently breastfeeding,” and “birth order” emerging as leading predictors.

Conclusions: These findings emphasize the importance of targeted intervention strategies that integrate all aspects of health systems, community dynamics, and household factors, contributing to data-driven approaches for improving under-five child survival in Southern Africa. While the RSF and DeepHit models are well-suited for survival analysis in contexts where the proportional hazards assumption is violated, the DeepFrailty model offers valuable interpretability by capturing unobserved heterogeneity, making it useful in settings where understanding latent community or household-level risk is critical.

Keywords: DeepFrailty, DeepHit, random survival forest, child mortality, variable importance.

Introduction

Under-five child mortality remains a significant global health concern, particularly in sub-Saharan Africa [1]. Addressing this critical issue requires high-quality data to guide individualised targeted interventions, aligning with international commitments such as the Sustainable Development Goals (SDGs) [2]. To support national policies and advance SDG 3 (Good Health and Well-being) and SDG 10 (to reduce inequalities to leave no child behind), Verhulst and colleagues [3] emphasised the importance of analysing under-five survival data by accounting not only for individual-level factors but also for communities and health system factors. This approach is crucial for identifying specific households or communities with high vulnerability and health systems with unmet needs, enabling the development of more effective resource allocation strategies for targeted child populations.

Clustered survival data are often observed in many settings. For example, in children’s health outcomes, an instance is the analysis of children who are clustered within same households or communities that have same healthcare systems. Such data often exhibit a hierarchical structure, with children nested within the same households, communities, health systems or geographical regions. These children may experience the event of interest throughout the follow-up period due to certain shared characteristics. The inherent correlation among children can be accounted for using a frailty term [4]. Children living within the same healthcare system, with similar access to medical care, or in the same geographical area may share unobserved characteristics, such as environmental factors. Analysing individual children’s data simultaneously within a hierarchical structure allows for direct adjustment of confounders, the

incorporation of proportional and non-proportional hazards in covariate effects and the observed and unobserved heterogeneity within and between children. [5–8].

Classical survival analysis methods have limitations when handling complex interactions between covariates in multilevel survival models, particularly when accounting for clustering and requiring strong parametric assumptions. The Random Survival Forest model, a machine learning approach, was initially proposed to fit survival data but they do not explicitly consider frailty modelling in the way traditional frailty models do [9]. More recently, two deep learning methods, DeepHit and DeepFrailty, have been developed to provide a general framework for modelling survival data. These methods effectively capture complex interactions and non-linear relationships among variables, uncovering hidden dependencies between risk factors and outcomes. The DeepFrailty model in particular explicitly handles the frailty modelling within the deep learning framework.

This study aims to predict under-five survival in Southern Africa by incorporating both individual- and household-level covariates. Its novelty lies in applying the DeepFrailty model to estimate mortality and predict survival while accounting for unobserved heterogeneity within and between communities. The model also captures interactions and nonlinear relationships. We emphasise the use of DeepFrailty predictive model, to support targeted interventions for under-five children. This approach improves the understanding of child survival determinants and informs data-driven public health policies.

This paper is arranged as follows. Section describes the Demographic Health Survey (DHS) datasets, the random survival forest, and two deep learning methods, DeepHit and DeepFrailty. Section describes the results. Section is the discussion and Section concludes the paper.

Materials and methods

Southern African countries DHS data

We aim to uncover hidden patterns within four national Demographic and Health Surveys (DHS) datasets from southern African countries, Malawi, South Africa, Zambia, and Zimbabwe. DHS datasets from these countries, are freely accessible, offer large, representative samples of under-five children, enabling the estimation of national child mortality hazards and survival probabilities [10]. In addition, we establish a rigorous methodological foundation for analysing child survival data in a region where child mortality is

often clustered within households or communities [11–13].

We used data from the most recent DHS conducted in Malawi (DHS 2016), South Africa (DHS 2016), Zambia (DHS 2018) and Zimbabwe (DHS 2015). These nationally representative household surveys collect detailed information on demographic, health, and socioeconomic characteristics, including full birth histories from women aged 15 – 49. For this study, we focused on singleton live births occurring within five years prior to the date of the survey interview. Children with missing information on age at death or survival status, or survival time values that are greater than 59 months, were excluded from the analysis. We also excluded births where key covariates used in modelling were missing or could not be recoded consistently across countries. The initial pooled dataset contained approximately 35 353 live births in four countries. The final sample analysed included 26 597 children, 12 671 children from Malawi, 2 760 children from South Africa, 6 730 children from Zambia, and 4 436 children from Zimbabwe. This is also summarised in figure 1 below.

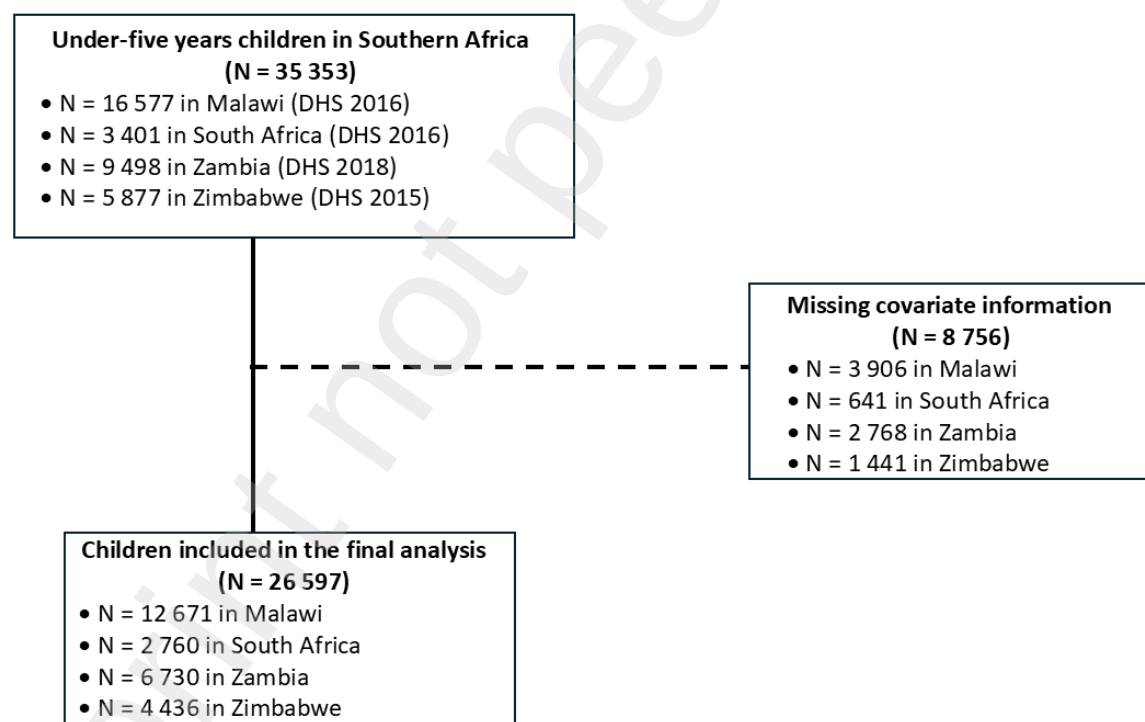


Fig 1. Flow diagram of under-five children in Malawi, South Africa, Zambia and Zimbabwe based on DHS surveys data.

The DHS surveys in these countries follow a standardised two-stage cluster sampling design. Enumeration areas are first selected from national census data, followed by systematic sampling of households within each cluster. In each selected household, eligible women aged 15 – 49 years provide retrospective child

birth history information, enabling the estimation of child survival (mortality). These surveys also include household characteristics, maternal health outcomes, and child-specific factors, allowing for analysis of survival determinants [10].

Data sets were obtained through the publicly accessible DHS programme repository and analysed to identify key predictors of survival for under-five years of age.

Study variables

This study employed a retrospective cohort design using cross-sectional data from the DHS. Survival analysis was enabled through the reconstruction of individual time-to-event data based on reported birth histories. The starting point was defined as the date of each child's birth. Children were followed from birth until the earliest of the following events, death before the age of five which is defined as the event and age five if alive, or the date of the survey interview, which is defined as censored. The follow-up period is therefore, that from birth up to a maximum of 59 months, depending on the age of the child at the time of the survey. The outcome variable is therefore, time from birth to death or censoring of a child before they celebrate their fifth birth day. All data were collected during the 2015 – 2016 survey period. For this analysis, we considered the following individual- and household-level factors as key predictors as under-five survival: (i) individual-level characteristics such as sex of child; (ii) household characteristics such as mother's age at first birth, mother's and father's education, birth order, number of births in the past five years, total number of children born, number of living children, mother's current marital status, wealth status, religion, post-partum amenorrhoea, current pregnant, current breastfeeding (iii) community-level and healthcare system characteristics such as drinking water source and delivery care.

Random survival forest

In the field of machine learning, an effective strategy to overcome the assumptions about proportional hazards associated with the Cox proportional hazards model is to use a random survival forest. Relaxing the assumption of proportional hazards allows the investigation of whether the effect of a covariate changes over time. The random forest method is characterised by its use of tree-based techniques to create an ensemble estimate for the cumulative hazard function. Leveraging ensemble methods with base learners like trees has the capacity to substantially improve predictive performance.

Fundamentally, the random survival forest method involves building a random forest, where mostly the log-rank test serves as the criterion for partitioning the data. Cumulative hazards or survival probabilities for leaf nodes within each tree are calculated and then aggregated across the ensemble to obtain cumulative hazard or survival probabilities. The Nelson-Aalen estimator is used to estimate the cumulative hazard function for each terminal node. The ensemble cumulative hazard function is then obtained by averaging over all trees in the forest. Similarly, the ensemble survival probabilities are derived by using the Kaplan-Meier estimator for the survival function. This approach enables the model to make dynamic survival predictions without relying on proportional hazards assumptions. The expansion of each tree adheres to the requirement that every terminal node contains a predefined minimum number of instances of the event of interest [14].

To assess the relative contribution of covariates to under-five mortality predictions, we computed Permutation Feature Importance (PFI) [15] scores using the Random Survival Forest (RSF) model. RSF was selected for this task due to its strong predictive performance and suitability for handling high-dimensional and categorical data. PFI estimates were obtained by measuring the decrease in the model's concordance index when each feature was randomly permuted. This method provides a model-agnostic and interpretable ranking of feature importance based on the impact of each variable on survival prediction accuracy.

Deep learning survival method

In recent years, there has been a substantial surge of research at the intersection of machine learning and survival analysis, particularly in the integration of neural networks [16]. As deep learning technologies and computational capabilities have evolved, remarkable achievements have been attained, enabling the application of various architectures to large datasets with diverse underlying processes and enhanced individualised learning.

Distinct clusters of methods within the domain of deep learning and survival analysis can be identified [16]. The first cluster revolves around the advancement of the fundamental Cox proportional hazards model. Some instances of these methods include DeepSurv [17] which is an extension of the Cox proportional hazards model to handle nonlinear relationships between the log-hazard and the features. Another approach explores fully parametric survival models that leverage Recurrent Neural Networks (RNNs) to

sequentially predict a distribution describing the time until the next event. Examples of such models include RNN-SURV [18], Weibull time-to-event RNN [19], and various others. Furthermore, there exist cutting edge deep learning neural networks, like DeepHit [16], effectively accommodating both competing risks scenarios and single-event analyses. The DeepHit model is an extension of the traditional Cox framework, allowing for more flexible modelling approaches beyond proportional hazards assumptions.

DeepHit

The DeepHit model utilises a neural network with feed-forward architecture, as suggested by [16] for survival analysis as shown in Figure 2.

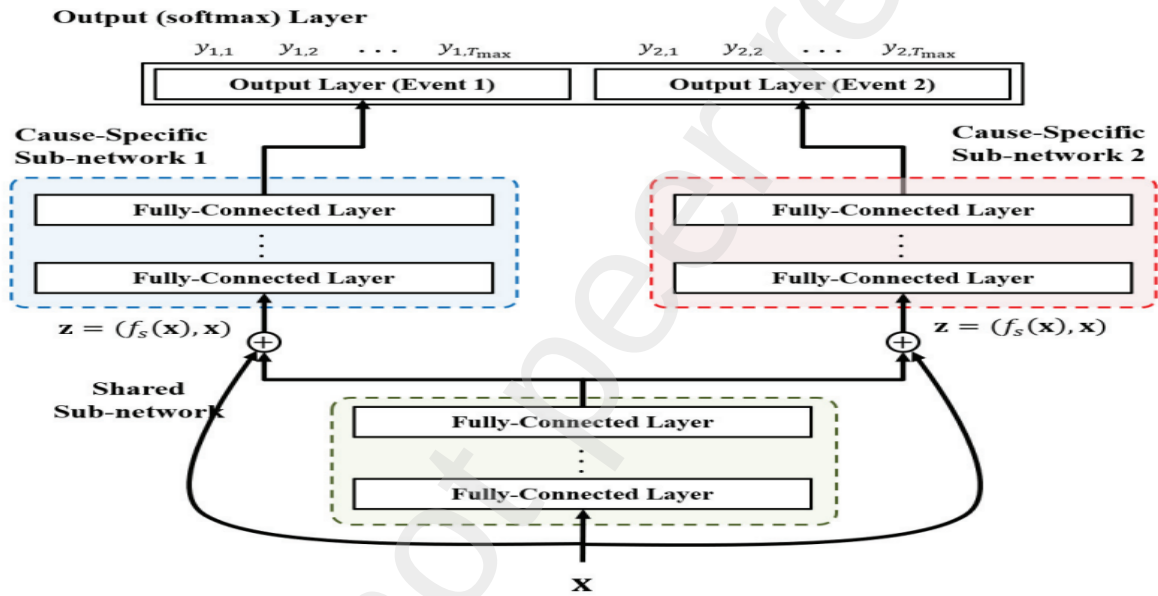


Fig 2. Feed-forward neural network architecture of the DeepHit model used for survival analysis [16].

Unlike the Cox proportional hazards model and the DeepSurv model, the DeepHit model does not operate within the proportional hazards framework, making it suitable for situations where the hazards ratio changes over time [14]. Moreover, the DeepHit model is also applicable to nonindependent events, making it a versatile model. The architecture of the DeepHit consists of several fully-connected layers, which are activated by a softmax function to produce a vector of estimated probabilities at different discrete times [16]. This flexibility made the DeepHit model a preferred choice for the analysis in this study because of the complexity of the datasets in question.

We employed the DeepHit model to predict under-five mortality risk using DHS data and a single event version of this model as implemented in the pycox package, constructing a feedforward neural network with two hidden layers, each followed by ReLU activation and dropout for regularization. The output

layer maps to a discretized time grid, enabling the model to learn the discrete probability mass function over survival time intervals. See table 1 for more of the model training details. The model was trained using early stopping, and we reported survival survival curves for under-five mortality data across the four Sub-Saharan African countries

Table 1. Hyperparameters for DeepHit model used in this study

Hyperparameter	Value
Architecture	2 hidden layers: [32, 32] nodes
Activation Function	ReLU
Dropout Rate	0.1
Number of events	1
Number of Time Intervals	100
Optimizer	Adam
Learning Rate	0.001
Loss Function	DeepHit loss, likelihood + ranking loss
Alpha (likelihood weight)	0.2
Sigma (ranking loss bandwidth)	0.1
Batch Size	256
Epochs	20 with early stopping
Evaluation Metrics	C-index, IBS

Neural frailty machine or DeepFrailty

The machine learning and deep learning models we have explored so far do not incorporate the concept of frailty. The Neural Frailty Machine (NFM) framework adopts the classical concept of multiplicative frailty from survival analysis within the proportional hazards framework, which is used to account for unobserved differences among individuals [20]. The NFM framework incorporates unobserved heterogeneity into neural survival regression models, including many commonly used survival regression models as special cases. This model was adopted in this study to account for the the unobserved heterogeneity which is very common in datasets on children health.

We employed a neural network model, we referred to as DeepFrailty, to predict under-five mortality risk using DHS data. The DeepFrailty model consists of a feedforward neural network with two hidden layers, the first maps the input to 128 hidden units using a ReLU activation and dropout regularization, and the second projects to a scalar log-risk score. A frailty component is captured through a monotonic transformation layer based on the Integrated Generalized Gamma (IGG) distribution, which ensures the

output is a valid survival function. The model was trained using the Monotone Negative Log-Likelihood loss, optimized with the Adam algorithm. Five-fold cross-validation was used, with model selection based on the minimum validation loss. Evaluation was conducted using the concordance index (C-index) and Integrated Brier Score (IBS). See table 2 for more of the details.

Table 2. Hyperparameters for DeepFrailty model used in this study

Hyperparameter	Value
Input Features	924 (one-hot encoded categorical)
Architecture	2 hidden layers: [128, 128] nodes
Activation Function	ReLU
Dropout Rate	0.5
Output	Scalar log-risk score
Monotonic Layer Hidden Units	128
Frailty Term	Integrated Generalized Gamma (IGG)
Positive Transformation	Exponential
Optimizer	Adam
Learning Rate	0.001
Loss Function	Monotone Negative Log-Likelihood
Weight Decay	0.01
Batch Size	256
Epochs	30
Evaluation Metrics	C-index, IBS

The datasets were preprocessed to include 26 variables, such as maternal age group, source of drinking water, birth order, education levels, and household characteristics. These covariates were one-hot encoded, resulting in a total of 924 input features. Missing values were excluded from the analysis. The outcome variable was time to event (in months), where the event was defined as child death before the age of five. Censoring was indicated by a status variable.

Measures of model performance

C-Index

In the field of survival analysis, the concordance index (C-index), also referred to as the concordance probability as discussed by [21], is the most commonly employed metric for evaluating the predictive capabilities of survival models. This metric measures the model's capacity to accurately rank pairs of survival times for distinct subjects. A higher C-index indicates better predictive accuracy [22].

The C-index is computed by evaluating the fraction of concordant pairs among all pairs of subjects. Two

subjects are considered concordant if the one who experiences the event of interest earlier also has a higher predicted risk compared to the one who experiences the event later. The C-index has a scale ranging from 0.5, suggesting random prediction to 1.0 which indicates a perfect prediction.

Brier score

It is used to measure the overall performance of the survival model. The advantage of the Brier score is that it is a proper scoring rule that addresses both calibration and discrimination [23]. It is calculated as the mean squared difference between the observed survival status and the predicted survival probability. Specifically, at a given time point t , the Brier score for an individual is determined by squaring the difference between the observed event outcome and the model's predicted probability of survival up to t . A Brier score below 0.25 indicates a well-calibrated and predictive model.

Ethics statement

The ethical statement for DHS dataset is available on the DHS ethical clearance certificate, which states that: The Internal Review Boards approved procedures for DHS public-use datasets do not in any way allow respondents, households, or sample communities to be identified. There are no names of individuals or household addresses in the data files. The geographic identifiers only go down to the regional level (where regions are typically very large geographical areas encompassing several states/provinces). Each enumeration area, Primary Sampling Unit (PSU) has a PSU number in the data file, but the PSU numbers do not have any labels to indicate their names or locations. In surveys that collect GIS coordinates in the field, the coordinates are only for the enumeration area as a whole, and not for individual households, and the measured coordinates are randomly displaced within a large geographic area so that specific enumeration areas cannot be identified.

Results

Table 3 presents key characteristics of child and maternal populations across four countries: Malawi, South Africa, Zambia, and Zimbabwe. It includes total number of children included for each country and compares the distribution of child sex, child age, vital status, maternal age at first birth, and parental education. Additional variables reported are access to drinking water, birth order, number of births in

Table 3. Characteristics of children from Malawi, South Africa, Zambia and Zimbabwe.

Characteristic	Malawi N = 16577	South Africa N = 3401	Zambia N = 9498	Zimbabwe N = 5877
Sex of child				
Male	8293/16577 (50.0%)	1752/3401 (52.0%)	4705/9498 (50.0%)	2889/5877 (49.0%)
Female	8284/16577 (50.0%)	1649/3401 (48.0%)	4793/9498 (50.0%)	2988/5877 (51.0%)
Child vital status				
Alive	16211/16577 (98.0%)	3348/3401 (98.0%)	9292/9498 (98.0%)	5713/5877 (97.0%)
Dead	366/16577 (2.2%)	53/3401 (1.6.0%)	206/9498 (2.2%)	164/5877 (2.8.0%)
Mother's age at first birth				
15 – 20 years	11554/16577 (70.0%)	1573/3401 (46.0%)	6582/9498 (69.0%)	3322/5877 (57.0%)
20 – 29 years	4916/16577 (30.0%)	1720/3401 (51.0%)	2829/9498 (30.0%)	2481/5877 (42.0%)
30 – 45 years	107/16577 (0.6%)	108/3401 (3.2%)	87/9498 (0.9%)	74/5877 (1.3%)
Mother's education				
None	11501/16577 (69.0%)	242/3401 (7.1.0%)	4483/9498 (47.0%)	758/5877 (13.0%)
Complete primary	3937/16577 (24.0%)	1885/3401 (55.0%)	3999/9498 (42.0%)	4688/5877 (80.0%)
Complete secondary	861/16577 (5.2.0%)	935/3401 (27.0%)	627/9498 (6.6%)	75/5877 (1.3%)
Higher Education	278/16577 (1.7%)	339/3401 (10.0%)	389/9498 (4.1%)	356/5877 (6.1%)
Father's education				
None	7047/13666 (52.0%)	169/1362 (12.0%)	2042/7057 (29.0%)	397/4937 (8.0%)
Complete primary	4081/13666 (30.0%)	563/1362 (41.0%)	3446/7057 (49.0%)	3856/4937 (78.0%)
Complete secondary	1923/13666 (14.0%)	458/1362 (34.0%)	999/7057 (14.0%)	202/4937 (4.1.0%)
Higher education	615/13666 (4.5%)	172/1362 (13.0%)	570/7057 (8.1%)	482/4937 (9.8%)
Drinking water				
Borehole	9925/16577 (60.0%)	113/3401 (3.3%)	2457/9498 (26.0%)	1476/5877 (25.0%)
Piped water	3657/16577 (22.0%)	2693/3401 (79.0%)	2099/9498 (22.0%)	2054/5877 (35.0%)
Surface/rain/pond/lake/tank	960/16577 (5.8%)	325/3401 (9.6%)	1388/9498 (15.0%)	670/5877 (11.0%)
Well	1897/16577 (11.0%)	94/3401 (2.8%)	3326/9498 (35.0%)	1411/5877 (24.0%)
Other	138/16577 (0.8%)	176/3401 (5.2%)	228/9498 (2.4%)	266/5877 (4.5%)
Birth order number				
1 child	4185/16577 (25.0%)	1229/3401 (36.0%)	2370/9498 (25.0%)	1613/5877 (27.0%)
2 – 3 children	6153/16577 (37.0%)	1682/3401 (49.0%)	3228/9498 (34.0%)	2692/5877 (46.0%)
4 – 6 children	4889/16577 (29.0%)	439/3401 (13.0%)	2736/9498 (29.0%)	1383/5877 (24.0%)
7+ children	1350/16577 (8.1%)	51/3401 (1.5%)	1164/9498 (12.0%)	189/5877 (3.2%)
Births in past 5 years				
1 child	9628/16577 (58.0%)	2479/3401 (73.0%)	4864/9498 (51.0%)	3544/5877 (60.0%)
2 children	6322/16577 (38.0%)	853/3401 (25.0%)	4033/9498 (42.0%)	2066/5877 (35.0%)
3 children	585/16577 (3.5%)	63/3401 (1.9%)	567/9498 (6.0%)	255/5877 (4.3%)
4+ children	42/16577 (0.3%)	6/3401 (0.2%)	34/9498 (0.4%)	12/5877 (0.2%)
Total number of children born				
1 child	3095/16577 (19.0%)	1025/3401 (30.0%)	1731/9498 (18.0%)	1194/5877 (20.0%)
2 children	3678/16577 (22.0%)	1113/3401 (33.0%)	1879/9498 (20.0%)	1597/5877 (27.0%)
3 children	2949/16577 (18.0%)	694/3401 (20.0%)	1557/9498 (16.0%)	1273/5877 (22.0%)
4+ children	6855/16577 (41.0%)	569/3401 (17.0%)	4331/9498 (46.0%)	1813/5877 (31.0%)
Current breastfeeding				
Not	7971/16577 (48.0%)	2512/3401 (74.0%)	4846/9498 (51.0%)	3492/5877 (59.0%)
Yes	8606/16577 (52.0%)	889/3401 (26.0%)	4652/9498 (49.0%)	2385/5877 (41.0%)
Current pregnant				
No	15515/16577 (94.0%)	3300/3401 (97.0%)	8671/9498 (91.0%)	5543/5877 (94.0%)
Yes	1062/16577 (6.4%)	101/3401 (3.0%)	827/9498 (8.7%)	334/5877 (5.7%)
Postpartum amenorrheic				
No	11580/16577 (70.0%)	2713/3401 (80.0%)	6694/9498 (70.0%)	4286/5877 (73.0%)
Yes	4997/16577 (30.0%)	688/3401 (20.0%)	2804/9498 (30.0%)	1591/5877 (27.0%)
Delivery care				
No	13081/16577 (79.0%)	2385/3401 (70.0%)	8856/9498 (93.0%)	5130/5877 (87.0%)
Yes	3496/16577 (21.0%)	1016/3401 (30.0%)	642/9498 (6.8%)	747/5877 (13.0%)
Number of living children				
1 child	3410/16524 (21.0%)	1085/3387 (32.0%)	1876/9464 (20.0%)	1350/5865 (23.0%)
2 children	3837/16524 (23.0%)	1136/3387 (34.0%)	1981/9464 (21.0%)	1709/5865 (29.0%)
3 children	3159/16524 (19.0%)	683/3387 (20.0%)	1609/9464 (17.0%)	1262/5865 (22.0%)
4+ children	6118/16524 (37.0%)	483/3387 (14.0%)	3998/9464 (42.0%)	1544/5865 (26.0%)
Religion				
Catholic	2196/16577 (13.0%)	0/3401 (0.0%)	7806/9498 (82.0%)	280/5877 (4.8%)
Muslim	83/16577 (0.5%)	0/3401 (0.0%)	0/9498 (0.0%)	16/5877 (0.3%)
Other Christians	14273/16577 (86.0%)	0/3401 (0.0%)	1606/9498 (17.0%)	5233/5877 (89.0%)
Others	25/16577 (0.2%)	3401/3401 (100.0%)	86/9498 (0.9%)	348/5877 (5.9%)
Mother's current marital status				
Never in union	574/16577 (3.5%)	1839/3401 (54.0%)	1082/9498 (11.0%)	304/5877 (5.2%)
Currently in union/living with a man	13841/16577 (83.0%)	1410/3401 (41.0%)	7349/9498 (77.0%)	4995/5877 (85.0%)
Formerly in union/living with a man	2162/16577 (13.0%)	152/3401 (4.5%)	1067/9498 (11.0%)	578/5877 (9.8%)
Wealth status				
Poorest	3733/16577 (23.0%)	813/3401 (24.0%)	2676/9498 (28.0%)	1191/5877 (20.0%)
Poorer	3589/16577 (22.0%)	844/3401 (25.0%)	2299/9498 (24.0%)	1023/5877 (17.0%)
Middle	3244/16577 (20.0%)	784/3401 (23.0%)	1869/9498 (20.0%)	918/5877 (16.0%)
Richer	3059/16577 (18.0%)	613/3401 (18.0%)	1418/9498 (15.0%)	1536/5877 (26.0%)
Richest	2952/16577 (18.0%)	347/3401 (10.0%)	1236/9498 (13.0%)	1209/5877 (21.0%)

the past five years, total number of children born, breastfeeding status, and current pregnancy status of mothers. Notable findings show variations in maternal education and access to drinking water across the countries. For instance, while borehole water is common in Malawi (60%), piped water dominates in South Africa (79%). Maternal age at first birth and total number of children born also differ, reflecting demographic and health system disparities across these four countries. The overall survival rate for the combined sample of 35,353 children from Zimbabwe, Malawi, Zambia, and South Africa, was 98%, defined as the proportion of children who survived beyond their fifth birthday.

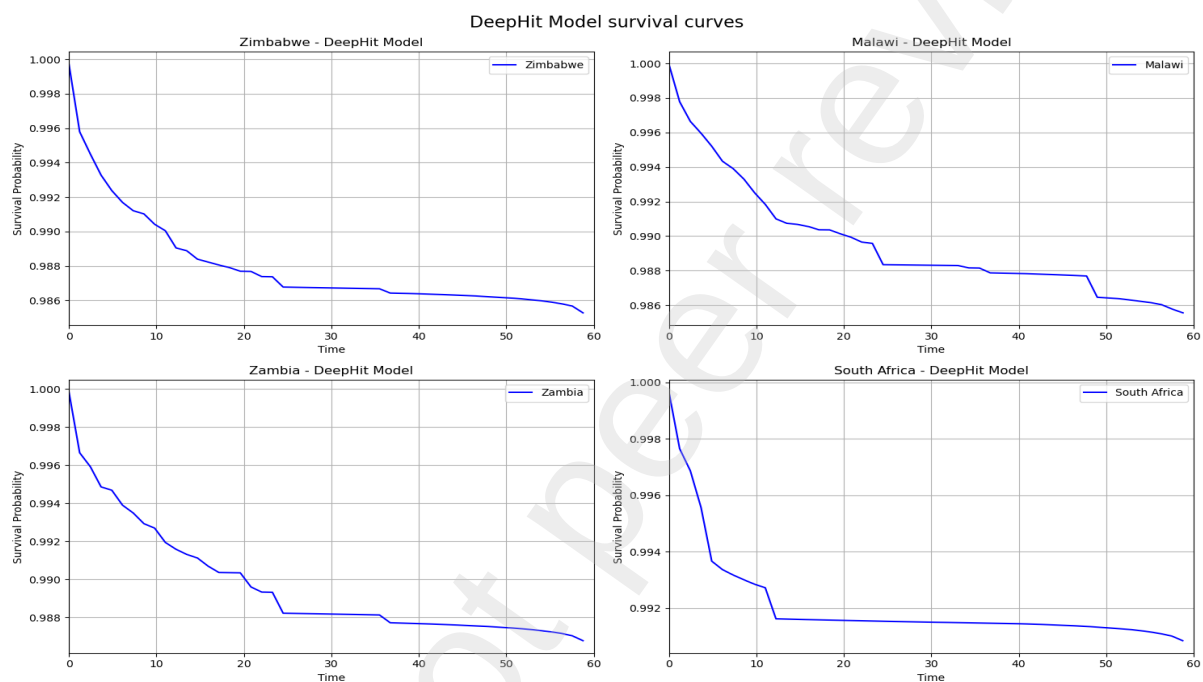


Fig 3. Survival curves for the children under the age of five in Zimbabwe, Malawi, Zambia and South Africa, respectively, extracted from the DeepHit model.

Fig 3 shows the DeepHit model's survival curves for Zimbabwe, Malawi, Zambia, and South Africa. The survival probability decreases more noticeably within the first 20 – 30 months before stabilizing, yet it remains above 0.98 for most of the observed period. This indicates a relatively high survival probability and relatively high survival.

Fig 4 shows the survival curves of the random survival forest model for Zimbabwe, Malawi, Zambia, and South Africa. Survival probabilities remain above 0.98 throughout the observed period. The random survival forest model shows a steeper initial decline but stabilizes quickly, indicating a moderate but consistent survival probability across these countries.

Fig 5 shows the survival curves generated from the DeepFrailty model for Zimbabwe, Malawi, Zambia, and South Africa. The curves appear similar across the four countries, with slight variations in the rates of de-

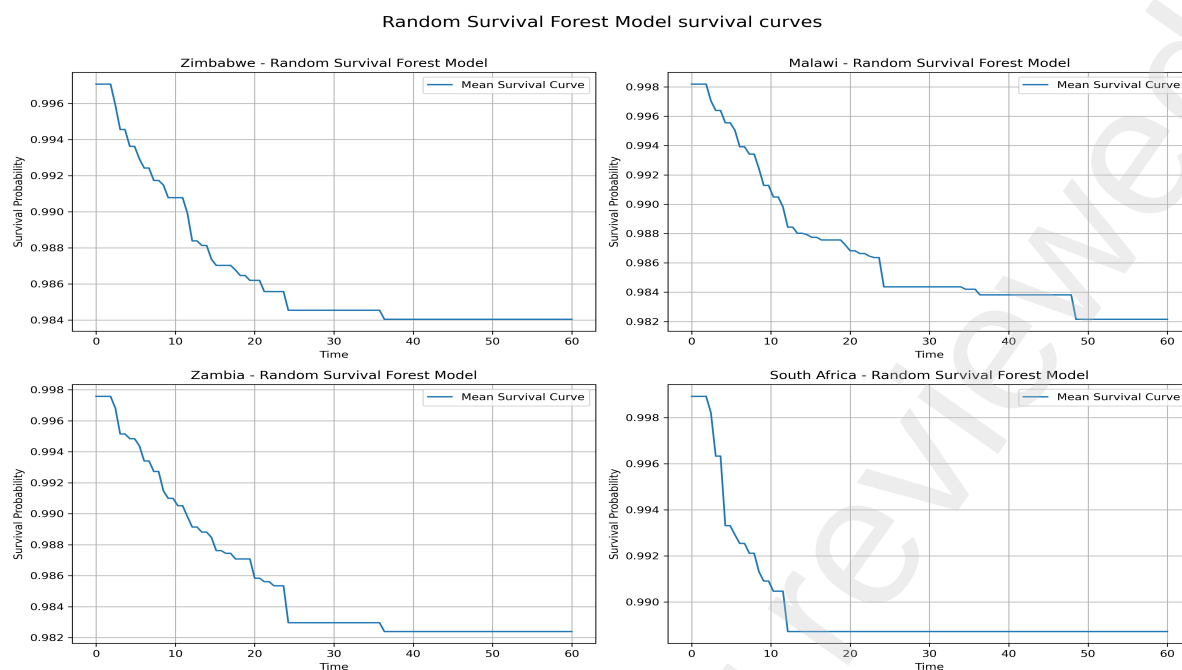


Fig 4. Survival curves for the children under the age of five in each country extracted from the random survival forest model.

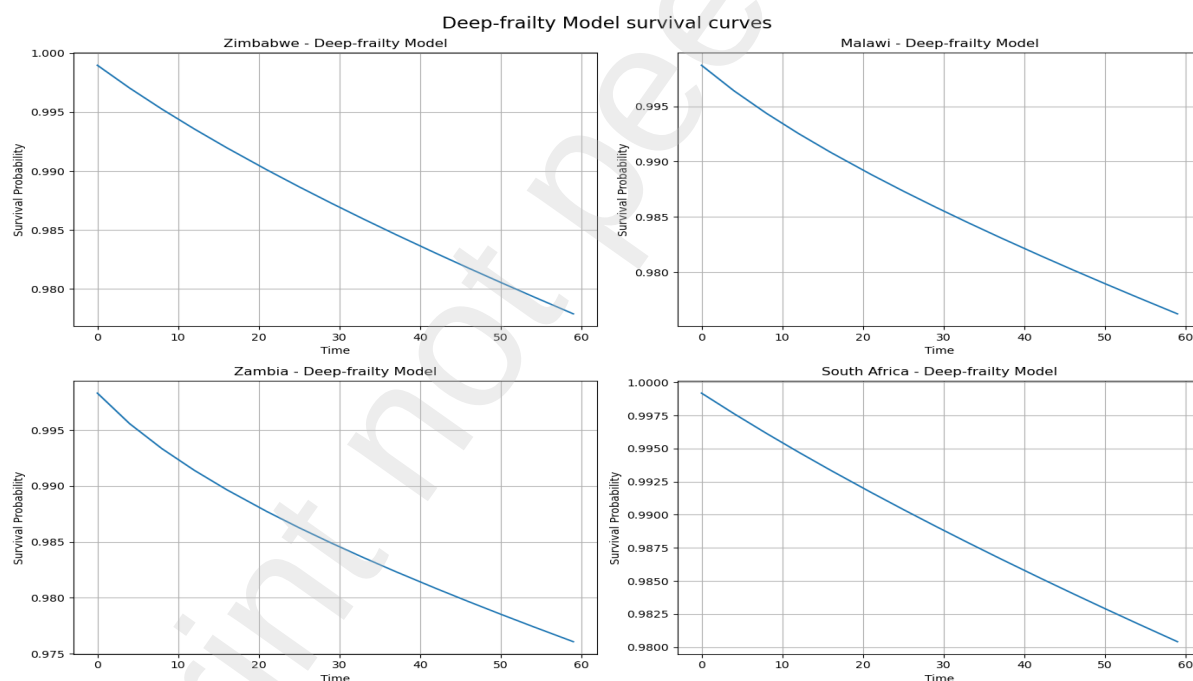


Fig 5. Survival curves for the children under the age of five in each country extracted from the DeepFrailty model.

cline, highlighting possible differences in survival patterns specific to each country's context. The survival probabilities remain above 0.98, indicating good survival under this model. This suggests therefore that, according to the DeepFrailty model, children under-five in these countries have a high likelihood of surviving.

Fig 6 illustrates the performance of the random survival forest, DeepHit, and DeepFrailty models for under-five mortality across Zimbabwe, Malawi, Zambia, and South Africa. The random survival forest model

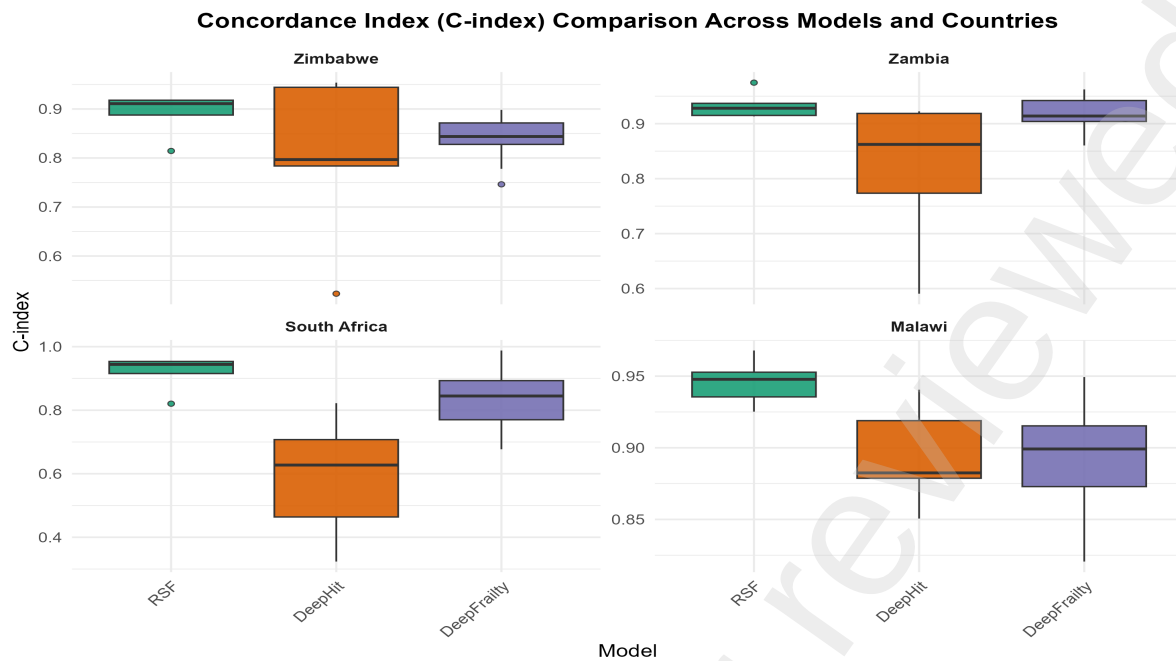


Fig 6. Comparison of concordance across random survival forest (random survival forest), DeepHit, and DeepFrailty models for Zimbabwe, Malawi, Zambia, and South Africa.

consistently achieves high C-index values across all countries, highlighting its strength in predicting survival probabilities and predicting which children are at higher relative risk of mortality. For the DeepHit model, the concordance index was 0.8003 [95% CI: 0.5840, 1.0000] in Zimbabwe, 0.8943 [95% CI: 0.8502, 0.9383] in Malawi, 0.8135 [95% CI: 0.6418, 0.9853] in Zambia, and 0.5889 [95% CI: 0.3437, 0.8341] in South Africa. For the RSF model, the concordance index was 0.8890 [95% CI: 0.8352, 0.9444] in Zimbabwe, 0.9458 [95% CI: 0.9255, 0.9661] in Malawi, 0.9336 [95% CI: 0.9027, 0.9646] in Zambia, and 0.9177 [95% CI: 0.8475, 0.9879] in South Africa. For the DeepFrailty model, the concordance index was 0.8417 [95% CI: 0.8260, 0.8574] in Zimbabwe, 0.8937 [95% CI: 0.8799, 0.9075] in Malawi, 0.9159 [95% CI: 0.9035, 0.9284] in Zambia, and 0.8385 [95% CI: 0.8039, 0.8730] in South Africa.

Comparing models within each country, we observe from the results that, in Zimbabwe, the confidence interval for DeepFrailty does not overlap with those of DeepHit or RSF, suggesting that both DeepHit and RSF significantly outperform DeepFrailty. There is no significant difference between DeepHit and RSF. In Malawi, DeepFrailty's interval slightly overlaps with that of DeepHit but not with RSF, indicating that RSF significantly outperforms DeepFrailty, while DeepHit may also offer improved performance, although the difference is less pronounced. No significant difference is observed between DeepHit and RSF. In Zambia, the confidence interval for DeepFrailty overlaps with both DeepHit and RSF, indicating no significant difference among the three models in this setting. In South Africa, DeepFrailty's interval

overlaps with both DeepHit and RSF, however, DeepHit and RSF do not overlap, suggesting a significant difference between them, with RSF outperforming DeepHit.

The DeepFrailty model, while exhibiting slightly lower C-index values compared to random survival forest, still performs acceptably (above 0.6), particularly in Zimbabwe and Zambia. Note that the inclusion of the frailty term in DeepFrailty allows it to model unobserved heterogeneity, which may partially explain underlying, unmeasured factors affecting mortality. This added benefit of addressing complex, unmeasured variability justifies its use in our study where such factors exist and influence outcomes.



Fig 7. Comparison of Integrated Brier Scores (IBS) across random survival forest, DeepHit, and DeepFrailty models for Zimbabwe, Malawi, Zambia, and South Africa

Fig 7 also shows the performance of the three models in the four countries based on the integrated Brier scores.

For the DeepHit model, the mean Brier score at time 30 months was 0.0140 [95% CI: 0.0088, 0.0193] in Zimbabwe, 0.0104 [95% CI: 0.0095, 0.0112] in Malawi, 0.0110, CI: [0.0089, 0.0132] in Zambia, and 0.0085 [95% CI: 0.0045, 0.0125] in South Africa. For the RSF model, the mean Brier score at time 30 months was 0.0097 [95% CI: 0.0052, 0.0141] in Zimbabwe, 0.0095 [95% CI: 0.0079, 0.0111] in Malawi, 0.0096 [95% CI: 0.0055, 0.0138] in Zambia, and 0.0062 [95% CI: 0.0022, 0.0102] in South Africa. For the DeepFrailty model, the mean Brier score at time 30 months was 0.0209 [95% CI: 0.0188, 0.0231] in Zimbabwe, 0.0198 [95% CI: 0.0170, 0.0226] in Malawi, 0.0212 [95% CI: 0.0195, 0.0230] in Zambia, and 0.0164 [95% CI: 0.0125, 0.0203] in South Africa. Tables 4 and 5 gives a full summary of these results.

Comparing models within each country based on the 95% confidence intervals of the Brier scores, in Zimbabwe, the confidence intervals for all three models, DeepFrailty, DeepHit, and RSF overlap, suggesting no statistically significant differences in performance. In Malawi, DeepHit significantly outperforms DeepFrailty, as their confidence intervals do not overlap. The intervals for DeepHit and RSF overlap slightly, indicating no significant difference between the two. However, RSF and DeepFrailty also overlap, implying no statistically significant difference between those two models. In Zambia, DeepHit significantly outperforms DeepFrailty, as do RSF over DeepFrailty, with no overlap in either case. The confidence intervals for DeepHit and RSF overlap slightly, suggesting comparable performance. In South Africa, the confidence intervals for DeepHit and RSF overlap, while those for RSF and DeepFrailty do not. DeepHit and DeepFrailty touch at the boundary of their intervals, which suggests that while DeepHit and RSF may not differ significantly, RSF likely outperforms DeepFrailty.

Although DeepHit performs reasonably well, RSF consistently outperforms it in both discrimination (C-index) and calibration (Brier score) across all four countries, including Malawi and Zambia. We therefore select RSF as the best-performing model. Its tree-based structure not only offers greater stability but also provides interpretable insights through feature importance rankings. This makes RSF particularly valuable for identifying and communicating key risk factors associated with under-five mortality. In practical applications where model transparency and robustness are crucial, these advantages make RSF the most balanced and actionable solution overall. Moreover, the Random Survival Forest yields the lowest Integrated Brier Score (IBS) values, indicating superior predictive accuracy over time. This low IBS suggests that RSF consistently provides reliable survival predictions, aligning with the high C-index values observed in Fig. 6. In contrast, the DeepHit model shows moderate IBS values generally below 0.25 reflecting acceptable but slightly lower performance. The DeepFrailty model presents the highest IBS values, suggesting reduced predictive accuracy. It is important to note, however, that the DeepFrailty model operates under the proportional hazards assumption, which may limit its flexibility in capturing complex temporal patterns. Nevertheless, its frailty component helps account for unobserved heterogeneity in risk, offering an interpretability advantage in understanding individual-level survival variability.

The random survival forest has the lowest IBS values, indicating superior predictive accuracy over time. This low score suggests that random survival forest consistently provides accurate survival predictions,

Table 4. Comparison of Mean C-index and 95% Confidence Intervals across models and countries

Model	Country	Mean C-index	95% CI Lower	95% CI Upper
DeepFrailty	Zimbabwe	0.8417	0.8260	0.8574
	Zambia	0.9159	0.9035	0.9284
	South Africa	0.8385	0.8039	0.8730
	Malawi	0.8937	0.8799	0.9075
DeepHit	Zimbabwe	0.8003	0.5840	1.0000
	Zambia	0.8135	0.6418	0.9853
	South Africa	0.5889	0.3437	0.8341
	Malawi	0.8943	0.8502	0.9383
RSF	Zimbabwe	0.8890	0.8352	0.9444
	Zambia	0.9336	0.9027	0.9646
	South Africa	0.9177	0.8475	0.9879
	Malawi	0.9458	0.9255	0.9661

Table 5. Comparison of Mean Brier Score and 95% Confidence Intervals across models and countries

Model	Country	Mean Brier Score	95% CI Lower	95% CI Upper
DeepFrailty	Zimbabwe	0.0209	0.0188	0.0231
	Zambia	0.0212	0.0195	0.0229
	South Africa	0.0165	0.0125	0.0203
	Malawi	0.0198	0.0170	0.0226
DeepHit	Zimbabwe	0.0140	0.0088	0.0193
	Zambia	0.0110	0.0089	0.0132
	South Africa	0.0085	0.0045	0.0125
	Malawi	0.0104	0.0095	0.0112
RSF	Zimbabwe	0.0097	0.0052	0.0141
	Zambia	0.0096	0.0055	0.0138
	South Africa	0.0062	0.0022	0.0102
	Malawi	0.0095	0.0079	0.0111

which is similar to the high C-index observed in Fig. 6. The DeepHit model shows moderate IBS values, generally performing well with values below 0.25. However, the DeepFrailty model has the highest IBS values, suggesting lower predictive accuracy. It is important to note that the DeepFrailty model used in this study is designed within the proportional hazards framework, a restriction that could impact its predictive performance. However, this predictive performance is balanced by the ability of the frailty term to capture unobserved heterogeneity risk factors that can affect survival.

Fig 8 shows ranked feature importance for predicting under-five mortality across Zimbabwe, Malawi, Zambia, and South Africa. These rankings are based on PFI [15], which measures the decrease in model performance when each feature is randomly permuted, thus revealing its contribution to the model's predictive accuracy. Features such as "birth order", "total number of children ever born", "total number of births in the last five years", "currently breastfeeding", and "births in the past year" play a substantial role in under-five mortality predictions in Zimbabwe. In Zambia, features such as "birth order", "total

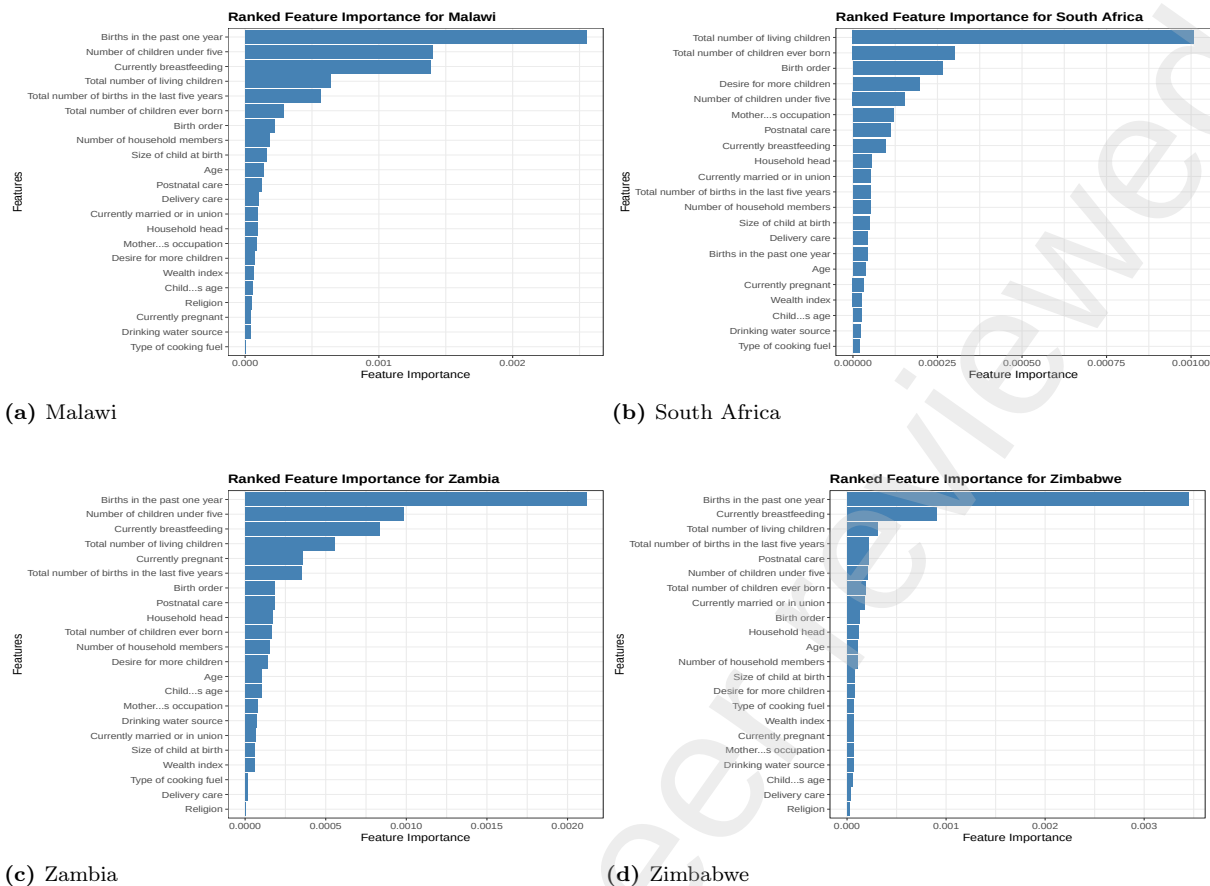


Fig 8. Ranked feature importance for (a) Malawi, (b) South Africa, (c) Zambia, and (d) Zimbabwe

number of children ever born”, “total number of births in the last five years”, and “currently breastfeeding” significantly impact child survival. For South Africa, “births in the past year” and “total number of births in the last five years” show high importance. “Currently breastfeeding” and “number of children under-five” are also among the features influencing survival probabilities for children under-five. In Malawi, the top features are “birth order”, “total number of children ever born”, “total number of births in the last five years”, “total number of living children”, “currently breastfeeding”, “number of children under-five”, and “births in the past year”.

Discussion

The results show that across all models and countries, survival probabilities remain above 0.97 for children under the age of five years, indicating that under-five children have high survival rates across Zimbabwe, Malawi, Zambia, and South Africa, a trend that has been observed in other research [5, 24]. This trend is encouraging and suggests that, despite regional differences, basic health and survival for young children remain stable. However, the variations between models emphasise the importance of considering different

survival factors unique to each country or region. In particular, the frailty model's performance provides a need to investigate unobserved heterogeneity factors further, which could include socioeconomic disparities or healthcare accessibility that impact survival in ways not directly observable (related to SDGs 3 and 10). The results also show how critical the role of household composition and maternal caregiving is in shaping under-five mortality outcomes across these countries. The consistent prominence of features such as; "total number of children ever born", "currently breastfeeding", "number of living children" and "birth order" suggests that family and caregiving dynamics are key survival determinants. For example, higher birth orders may reflect reduced parental resources or attention, increasing mortality risk for subsequent children. Similarly, breastfeeding remains a universally protective factor, enhancing survival chances in early childhood. Country-specific differences in the feature rankings highlight the need for targeted interventions that address unique risk factors in each context. For example, in Zimbabwe and Zambia, focusing on educating mothers about the importance of birth order and equitable resource allocation could be beneficial. In Malawi, supporting family planning and maternal health services may address the mortality risks associated with a larger family structure. In South Africa, interventions that support maternal health and emphasise breastfeeding could directly impact under-five mortality. The inclusion of a frailty term in the DeepFrailty model, despite slightly lower performance metrics, provides value by capturing unobserved heterogeneity variables that influence child survival, such as unmeasured socioeconomic or environmental conditions. Although the random survival forest model is the most accurate model across these four datasets, the DeepFrailty model's capacity to capture complex and non-linear relationships while addressing unobserved heterogeneity influencing survival makes it a valuable option to explore further in the DHS dataset analysis. However, incorporating interpretability is a further study setting that the DeepFrailty model cannot achieve at this stage.

It is important to note that in this study that, although DeepFrailty provides a flexible, semi-parametric framework for modeling survival functions through a monotonic neural network architecture, it did not consistently outperform the RSF model across all performance metrics in this study. One possible explanation could be in the structure of the RSF model, that it uses an ensemble of decision trees, enabling it to capture complex non-linear interactions and hierarchical structures in the data with minimal hyperparameter tuning. In contrast, DeepFrailty requires careful optimization in a high-dimensional parameter space and is more sensitive to architectural configurations, training dynamics, and regularization

strategies. Most critically, RSF is particularly well-suited to high-dimensional and sparse data such as the DHS datasets due to its tree based splitting mechanism. This allows it to more effectively identify and exploit latent clustering or heterogeneity at the community or household level, which may be less easily learned by neural network-based models without substantial tuning or data augmentation.

Conclusion

This study shows high survival probabilities for under-fives (> 0.97) in Zimbabwe, Malawi, Zambia, and South Africa, indicating positive health trends for the age group. The most important determinants of child survival highlighted by this study include household composition and maternal care giving, with characteristics such as “births in the past year”, “total number of living children”, “total children ever born”, “currently breastfeeding”, and “birth order”, which were consistently ranked at the top in influencing survival. Country-specific interventions, such as promoting family planning or breastfeeding initiatives, could further improve survival rates. While the random survival forest model had best predictive performance among the three models, the DeepFrailty model captured complex and unobserved heterogeneity factors like unobserved socioeconomic disparities at community level. Despite its lower performance as compared to the other models used in this study, its ability to address heterogeneity makes it valuable for further research. Incorporating the model’s interpretability remains a priority to better inform targeted, evidence-based interventions aligned with Sustainable Development Goals (SDGs 3 and 10).

Limitation of the study

Survival was consistently high in all four countries, with only minimal variation observed. This limited variability in the observed outcome and reduced the discriminative capacity of the models, particularly in distinguishing subtle risk differences between individuals. Missing covariate data commonly occur in epidemiological research, and this was a challenge we faced in this study. Future analyses should consider extending the multiple imputation substantive model compatible-fully conditional specification approach to account for missing data in this setting.

Acknowledgments

The authors gratefully acknowledge the London Mathematical Society for supporting this work through the Scheme 5 “Collaborations with Developing Countries” grant. We would like to thank Eric Ohuma for his assistance during the project execution and for helping to host JBN during her research visit at LSHTM. JBN, a Humboldt scholar, set up the software for the DeepFrailty model during her Humboldt research fellowship stay with the Emmy Noether Junior Research Group: Beyond Prediction, Statistical Inference with Machine Learning at the Leibniz Institute for Prevention Research and Epidemiology (BIPS) in Bremen, Germany, enabling authors to run and utilize the model.

Author contributions

All authors have contributed equally on conceptualisation, formal analysis, interpretation of results, in writing the original draft, review, and editing.

Funding

WCS is funded through a Cancer Research UK programme grant (EPNCZS34).

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